

Alberto Pozanco

PERSONAL DATA

NATIONALITY:	Spanish
DATE OF BIRTH:	3 March 1993
EMAIL:	alberto.pozanco@gmail.com
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RESEARCH INTERESTS

Artificial Intelligence • Automated Planning • Heuristic Search • Goal Reasoning Goal Recognition • Explainable AI • Multiagent Systems • Machine Learning

WORK EXPERIENCE

J.P. MORGAN 2020 - Currently	[2023-Currently]	AI Research Vice President (Madrid, Spain)
	[2022-2023]	AI Research Senior Associate (Madrid, Spain)
	[2021-2022]	AI Research Senior Associate (London, UK)
	[2020]	AI Research Intern (London, UK)
UNIVERSIDAD CARLOS III 2016 - 2021	[2019-2021]	Research and Teaching Assistant (Madrid, Spain)
	[2019]	Research Visitor at RMIT (Melbourne, Australia)
	[2016-2019]	Research and Teaching Assistant (Madrid, Spain)
JAPPYSAVER 2014	[2014]	Data Analyst (Madrid, Spain)

EDUCATION

MARCH 2021	PhD in Computer Science and Technology Universidad Carlos III, Madrid, Spain Thesis: "Goal Reasoning for Autonomous Agents using Automated Planning" GRADE: Cumlaude
SEPTEMBER 2016	MSc in Computer Science and Technology with specialization in Artificial Intelligence, Universidad Carlos III, Madrid, Spain Thesis: "Planning and Learning for Urban Traffic Control" GRADE: A
SEPTEMBER 2015	BSc in Computer Science with specialization in Artificial Intelligence, Universidad Carlos III, Madrid, Spain Thesis: "Analysis and Implementation of an Automatic Poker Player" GRADE: A

PUBLICATIONS

Conference Papers

1. **Computing Planning Centroids and Minimum Covering States using Symbolic Bidirectional Search.** Pozanco, A.; Torralba, A.; Borrajo, D. In *Proceedings of ICAPS'24*, 455-463, Banff (Canada), 2024
2. **Contrastive Explanations of Multi-agent Optimization Solutions.** Zehtabi, P.; Pozanco, A.; Bloch, A.; Borrajo, D.; Kraus, S. In *Proceedings of ICAPS'24*, 671-679, Banff (Canada), 2024
3. **Generalising Planning Environment Redesign.** Pozanco, A.; Fraga Pereira, R.; Borrajo, D. In *Proceedings of AAAI'24*, 20230-20237, Vancouver (Canada), 2024.
4. **Generating Replanning Goals through Multi-objective Optimization in Response to Execution Observation.** Pozanco, A.; Borrajo, D.; Veloso, M. In *Proceedings of ECAI'23*, 1898-1905, Krakow (Poland), 2023.
5. **Combining Heuristic Search and Linear Programming to Compute Realistic Financial Plans.** Pozanco, A.; Papasotiriou, K.; Borrajo, D.; Veloso, M. In *Proceedings of ICAPS'23*, 527-531, Prague (Czech Republic), 2023.
6. **Explaining Preference-driven Schedules: the EXPRES Framework.** Pozanco, A.; Mosca, F.; Zehtabi, P.; Magazzeni, D.; and Kraus, In *Proceedings of ICAPS'22*, 710-718, (Online), 2022.
7. **Multi-tier Automated Planning for Adaptive Behavior.** Ciolek, D; D'Ippolito, N; Pozanco, A; and Sardina, S. In *Proceedings of ICAPS'20*, 66-74, (Online), 2020.
8. **Finding Centroids and Minimum Covering States in Planning.** Pozanco, A; E-Martín, Y; Fernández, S; and Borrajo, D. In *Proceedings of ICAPS'19*, 348-352, Berkeley (USA), 2019.
9. **Error Analysis and Correction for Weighted A*'s Suboptimality.** Holte, R.C.; Majadas, R.; Pozanco, A.; and Borrajo, D. In *Proceedings of SoCS'19*, 135-139, Napa (USA), 2019.
10. **Counterplanning using Goal Recognition and Landmarks.** Pozanco, A; E-Martín, Y; Fernández, S; and Borrajo, D. In *Proceedings of IJCAI'18*, 4808-4814, Stockholm (Sweden), 2018.

Journal Articles

1. **Train Route Planning as a Multi-agent Path Finding Problem.** Salerno M., E-Martín Y., Fuentetaja R., Gragera A., Pozanco A., Borrajo D. In *Advances in Artificial Intelligence. CAEPIA 2021. Lecture Notes in Computer Science, vol 12882. Springer* https://doi.org/10.1007/978-3-030-85713-4_23, 2021
2. **On-line Modeling and Planning for Urban Traffic Control.** Pozanco, A; Fernández, S; and Borrajo, D. *Expert Systems*, <https://doi.org/10.1111/exsy.12693>, 2021.
3. **Learning-driven Goal Generation.** Pozanco, A; Fernández, S; and Borrajo, D. *AI Communications*, vol. 31, no. 2, pp. 137-150, 2018.

Workshop and Other Non-Archived Papers

1. **On Learning Action Costs from Input Plans.** Morales, M.; Pozanco, A.; Canonaco, G.; Gopalakrishnan, S.; Borrajo, D.; Veloso, M. In *arxiv preprint arXiv:2408.10889*, 2024
2. **TRIP-PAL: Travel Planning with Guarantees by Combining Large Language Models and Automated Planners.** de la Rosa, T.; Gopalakrishnan, S.; Pozanco, A.; Zeng, Z.; Borrajo, D. In *arxiv preprint arXiv:2406.10196*, 2024

3. **On the Sample Efficiency of Abstractions and Potential-Based Reward Shaping in Reinforcement Learning.** Canonaco, G.; Ardon, L.; Pozanco, A.; Borrajo, D. In *arxiv preprint arXiv:2404.07826*, 2024
4. **Can Soft Goals be Compiled Away More Efficiently?** Pozanco, A. In *Proceedings of WEEP Workshop, ICAPS'24*, Banff (Canada), 2024
5. **On Computing Plans with Uniform Action Costs.** Pozanco, A.; Borrajo, D.; Veloso, M. In *arxiv preprint arXiv:2402.09877* and *Proceedings of HAXP Workshop, ICAPS'24*, 2024
6. **Exploring the Limitations of using Large Language Models to Fix Planning Tasks.** Gragera, A.; Pozanco, A. In *Proceedings of KEPS Workshop, ICAPS'23*, Prague (Czech Republic), 2023
7. **Fairness in Multi-Agent Planning** Pozanco, A. and Borrajo, D. In *arxiv preprint arXiv:2212.00506*, 2022
8. **Inapplicable Actions Learning for Knowledge Transfer in Reinforcement Learning** Ardon, L.; Pozanco, A.; Borrajo, D.; and Ganesh, S. In *arxiv preprint arXiv:2211.15589* and *Proceedings of PRL Workshop, ICAPS'23*, 2023
9. **Anticipatory Counterplanning** Pozanco, A.; E-Martín, Y.; Fernández, S.; and Borrajo, D. In *arxiv preprint arXiv:2203.16171*, 2022
10. **Filtering Top-K Relevant Plans.** Salerno, M.; Tabernero, M.; Pozanco, A.; Fuentetaja, R. In *Proceedings of the 3rd Workshop on Planning for Financial Services (FinPlan), ICAPS'22*, Singapore (Singapore), 2022
11. **PFPT: a Personal Finance Planning Tool by means of Heuristic Search and Automated Planning.** Pozanco, A.; Papasotiriou, K.; Borrajo, D. In *Proceedings of the 3rd Workshop on Planning for Financial Services (FinPlan), ICAPS'22*, Singapore (Singapore), 2022
12. **Proving Security of Cryptographic Protocols using Automated Planning.** Pozanco, A.; Polychroniadou, A.; Magazzeni, D.; Borrajo, D. In *Proceedings of the 2nd Workshop on Planning for Financial Services (FinPlan), ICAPS'21*, Guangzhou (China), 2021
13. **A Planning Approach to Agile Project Management. The JIRA Planner.** Alami, S.; Zehetab, P.; Silva, R.; Pozanco, A.; Magazzeni, D.; Borrajo, D.; Shah, S.; Veloso, M. In *Proceedings of the 2nd Workshop on Planning for Financial Services (FinPlan), ICAPS'21*, Guangzhou (China), 2021
14. **Get me to Safety! Escaping from Risks using Automated Planning** Pozanco, A.; E-Martín, Y.; Fernández, S.; and Borrajo, D. In *Proceedings of IntEx/GR Workshop, ICAPS'20*, Nancy (France), 2020
15. **Insights from the 2018 IPC Benchmarks.** Cenamor, I.; and Pozanco, A. In *Proceedings of 5th Workshop on the International Planning Competition, ICAPS'19*, Berkeley (USA), 2019.
16. **Counterplanning in Real-Time Strategy Games through Goal Recognition.** Pozanco, A.; Blanco, A.; E-Martín, Y.; Fernández, S.; and Borrajo, D. In *Proceedings of 6th Workshop on Goal Reasoning, IJCAI'18*, Stockholm (Sweden), 2018.
17. **Distributed Planning and Model Learning for Urban Traffic Control.** Pozanco, A.; Fernández, S.; and Borrajo, D. In *Proceedings of Workshop on Knowledge Engineering for Planning and Scheduling, ICAPS'18*, Delft (Netherlands), 2018.
18. **Counterplanning using Goal Recognition and Landmarks** Pozanco, A.; E-Martín, Y.; Fernández, S.; and Borrajo, D. In *Proceedings of 6th Workshop on Distributed and Multi-Agent Planning, ICAPS'18*, Delft (Netherlands), 2018
19. **Urban Traffic Control Assisted by AI Planning and Relational Learning.** Pozanco, A.; Fernández, S.; and Borrajo, D. In *Proceedings of 9th International Workshop on Agents in Traffic and Transportation (IJCAI'16)*, New York (USA), 2016.

20. **On Learning Planning Goals for Traffic Control.** Pozanco, A.; Fernández, S.; and Borrajo, D. In *Proceedings of 4th Workshop on Goal Reasoning (IJCAI'16)*, New York (USA), 2016.

PATENTS

1. **Method and System for Solving Subset Sum Matching Problem Using Dynamic Programming Approach.** Wu, Y.; Zehtabi, P.; Pozanco, A.; Borrajo, D.; Magazzeni, D.; Veloso, M.; Cashmore, M.; Deng, J. *US Patent App. 18/208,608*, 2023.
2. **Method and System for Solving Subset Sum Matching Problem Using Search Approach.** Wu, Y.; Zehtabi, P.; Pozanco, A.; Borrajo, D.; Magazzeni, D.; Veloso, M.; Cashmore, M.; Deng, J. *US Patent App. 18/208,603*, 2023.
3. **Method and System for Personal Financial Planning by Artificial Intelligence Search.** Pozanco, A.; Papasotiriou, K.; Borrajo, D.; Stefanucci, A.; Marchessotti, N.; Staddon, J.; Veloso, M. *US Patent App. 17/939,433*, 2022.
4. **Method and System for Space Planning by Artificial Intelligence Reasoning.** Silva, R.; Pozanco, A.; Zehtabi, P.; Magazzeni, D.; Veloso, M. *US Patent App. 17/647,205*, 2022.
5. **Method and System for Optimization of Task Management Issue Planning.** Alami, S.; Pozanco, A.; Shah, S.; Magazzeni, D.; Borrajo, D.; Zehtabi, P.; Silva, R.; Veloso, M. *US Patent 11,681,963*, 2021.
6. **Method and System for Providing Dynamic Workspace Scheduler.** Zehtabi, S.; Pozanco, A.; Silva, R.; Alami, S.; Borrajo, D.; Mahfouz, M.; Magazzeni, D.; Veloso, M.; Rasco, T.; Horn, J.; Blackwell, A.; Herschmann, D. *US Patent App. 17/450,609*, 2021.

TEACHING

Teaching Assistant

- **Machine Learning**, 3th of BSc. in Data Science and Engineering, Spring 2020. Universidad Carlos III de Madrid. (*English*)
- **Machine Learning**, 3th of BSc. in Computer Science and Engineering, Spring 2020. Universidad Carlos III de Madrid. (*Spanish*)
- **Machine Learning**, 3th of BSc. in Computer Science and Engineering, Spring 2019. Universidad Carlos III de Madrid. (*Spanish*)
- **Artificial Intelligence in Games**, 4th of BSc. in Computer Science and Engineering, Fall 2018. Universidad Carlos III de Madrid. (*Spanish*)

Supervision of MSc. Thesis

- **Counterplanning in Real-Time Strategy Games through Goal Recognition**, by Alejandro Blanco (co-supervised with Susana Fernández), Universidad Carlos III de Madrid.

Membership in BSc. Thesis Committees

- October 2020, BSc. in Computer Science and Engineering, Universidad Carlos III de Madrid.
- July 2018, BSc. in Industrial Technology, Universidad Carlos III de Madrid.
- October 2018, BSc. in Computer Science and Engineering, Universidad Carlos III de Madrid.

SCIENTIFIC ACTIVITIES

Invited Talks and Seminars

- Talk on "AI Search and Planning" at Learn Data & Analytics week, J.P. Morgan, Online (2023)
- Talk on "AI in Financial Institutions" at Universidad Internacional de Valencia, Valencia, Spain (2022)
- Talk on "Proving Security of Cryptographic Protocols using Automated Planning" at Simons Institute Industry Day, Berkeley, USA (2021)
- Talk on "Goal Reasoning for Autonomous Agents using Automated Planning" at RMIT University, Melbourne, Australia (2019)

Workshop Organization

- Organizer of the 2nd edition of the Bridge on AI for Financial Services at AAAI'24, Vancouver (Canada), 2024
- Organizer of the 4th edition of the Planning and Scheduling for Financial Services Workshop (Finplan) at ICAPS'23, Prague (Czech Republic), 2023
- Organizer of the 3rd edition of the Planning for Financial Services Workshop (FinPlan) at ICAPS'22, Online

Conference Reviewer

- **Program Committee** of the *27th European Conference on Artificial Intelligence (ECAI'24)*.
- **Program Committee** of the *34th International Conference on Automated Planning and Scheduling (ICAPS'24)*.
- **Program Committee** of the *38th AAAI Conference on Artificial Intelligence (AAAI'24)*.
- **Program Committee** of the *26th European Conference on Artificial Intelligence (ECAI'23)*.
- **Program Committee** of the *32nd International Joint Conference on Artificial Intelligence (IJCAI'23)*.
- **Program Committee** of the *37th AAAI Conference on Artificial Intelligence (AAAI'23)*.
- **Program Committee** of the *31st International Joint Conference on Artificial Intelligence (IJCAI'22)*.
- **Additional Reviewer** of the *32nd International Conference on Automated Planning and Scheduling (ICAPS'22)*.
- **Program Committee** of the *30th International Joint Conference on Artificial Intelligence (IJCAI'21)*.
- **Additional Reviewer** of the *35th AAAI Conference on Artificial Intelligence (AAAI'21)*.
- **Additional Reviewer** of the *34th AAAI Conference on Artificial Intelligence (AAAI'20)*.
- **Additional Reviewer** of the *29th International Conference on Automated Planning and Scheduling (ICAPS'19)*.
- **Additional Reviewer** of the *33th AAAI Conference on Artificial Intelligence (AAAI'19)*.
- **Additional Reviewer** of the *28th International Conference on Automated Planning and Scheduling (ICAPS'18)*.

Journal Reviewer

- Artificial Intelligence Journal (AIJ), since 2022.
- Knowledge Engineer Review (KER), since 2023.

COMPUTER SKILLS

- **Programming Languages:** Python, C++, Java
- **Frameworks:** Scikit-learn, Numpy, Pandas, PyTorch, Dash